Lesson Planning

Complete Lesson Plan Book

Class: Class 11 – ICom

Subject: Business Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Number System and Sets

Unit 2: Quadratic Equations

Unit 3: Matrices and Determinants

Unit 4: Sequences and Series

Unit 5: Permutations and Combinations

Unit 6: Percentage, Profit and Loss

Unit 7: Simple and Compound Interest

Unit 8: Annuities: Present and Future Value

DM

Unit 1: Number System and Sets

Learning Objectives

• Classify numbers: natural, integer, rational, real.

• Perform operations on sets: union, intersection, complement.

• Solve business problems using Venn diagrams.

Key Concepts

• Number sets and their hierarchy.

• Set notation: roster and set-builder forms.

• Union, intersection, difference of sets.

• Venn diagrams for two and three sets.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Number System and Sets to activate prior knowledge.

2. Present the core ideas of Number System and Sets with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Number System and Sets.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Number System and Sets and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Number System and Sets for classroom display.

Assessment Questions

1. Classify the given numbers into sets.

2. Find the union and intersection of two sets.

3. Solve a survey problem using a Venn diagram.

Homework

• Write a one-page summary of Number System and Sets in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 2: Quadratic Equations

Learning Objectives

• Solve quadratic equations by factorisation.

• Apply the quadratic formula correctly.

• Solve word problems leading to quadratics.

Key Concepts

• Standard form: ax^2 + bx + c = 0.

• Factorisation method for simple quadratics.

• Quadratic formula: x = (-b +/- sqrt(b^2-4ac)) / 2a.

• Discriminant tells the nature of roots.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Quadratic Equations to activate prior knowledge.

2. Present the core ideas of Quadratic Equations with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Quadratic Equations.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Quadratic Equations and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Quadratic Equations for classroom display.

Assessment Questions

1. Solve x^2 - 5x + 6 = 0 by factorisation.

2. Solve 2x^2 + 3x - 2 = 0 using the formula.

3. Find the discriminant and nature of roots.

Homework

• Write a one-page summary of Quadratic Equations in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 3: Matrices and Determinants

Learning Objectives

• Add, subtract and multiply matrices.

• Find the determinant of 2x2 and 3x3 matrices.

• Solve two equations using Cramer's rule.

Key Concepts

• Matrix: rectangular array of numbers.

• Order, types: row, column, square, identity.

• Multiplication: row-by-column rule.

• Determinant and Cramer's rule for linear equations.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Matrices and Determinants to activate prior knowledge.

2. Present the core ideas of Matrices and Determinants with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Matrices and Determinants.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Matrices and Determinants and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Matrices and Determinants for classroom display.

Assessment Questions

1. Multiply the two given matrices.

2. Find the determinant of a 3x3 matrix.

3. Solve the equations by Cramer's rule.

Homework

• Write a one-page summary of Matrices and Determinants in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 4: Sequences and Series

Learning Objectives

• Identify arithmetic and geometric sequences.

• Find the nth term and sum of an AP.

• Find the nth term and sum of a GP.

Key Concepts

• Sequence vs series: terms vs their sum.

• AP: common difference; nth term a + (n-1)d.

• Sum of AP: n/2 [2a + (n-1)d].

• GP: common ratio; sum formulas for r<1 and r>1.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Sequences and Series to activate prior knowledge.

2. Present the core ideas of Sequences and Series with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Sequences and Series.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Sequences and Series and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Sequences and Series for classroom display.

Assessment Questions

1. Find the 10th term of the given AP.

2. Find the sum of the first 20 terms of an AP.

3. Find the sum of a GP with given first term and ratio.

Homework

• Write a one-page summary of Sequences and Series in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 5: Permutations and Combinations

Learning Objectives

• Distinguish permutations from combinations.

• Apply nPr and nCr formulas.

• Solve selection and arrangement problems.

Key Concepts

• Permutation: arrangement; order matters.

• Combination: selection; order does not matter.

• nPr = n!/(n-r)!; nCr = n!/(r!(n-r)!).

• Business uses: committees, teams, samples.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Permutations and Combinations to activate prior knowledge.

2. Present the core ideas of Permutations and Combinations with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Permutations and Combinations.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Permutations and Combinations and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Permutations and Combinations for classroom display.

Assessment Questions

1. In how many ways can 3 books be arranged from 7?

2. How many committees of 4 can be formed from 9 persons?

3. State the difference between permutation and combination.

Homework

• Write a one-page summary of Permutations and Combinations in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 6: Percentage, Profit and Loss

Learning Objectives

• Convert between fractions, decimals and percentages.

• Calculate profit, loss and their percentages.

• Find discount, commission and marked price.

Key Concepts

• Percentage as parts per hundred.

• Profit% and loss% always on cost price.

• Discount on marked price; commission on sales.

• Successive discounts and their single equivalent.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Percentage, Profit and Loss to activate prior knowledge.

2. Present the core ideas of Percentage, Profit and Loss with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Percentage, Profit and Loss.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Percentage, Profit and Loss and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Percentage, Profit and Loss for classroom display.

Assessment Questions

1. A shopkeeper buys for Rs. 800 and sells for Rs. 960. Find profit%.

2. Find the single discount equal to 20% and 10%.

3. An agent gets 5% commission on Rs. 40,000 sales. Find it.

Homework

• Write a one-page summary of Percentage, Profit and Loss in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 7: Simple and Compound Interest

Learning Objectives

• Calculate simple interest and amount.

• Calculate compound interest and amount.

• Compare simple and compound interest.

Key Concepts

• SI = P x R x T / 100; amount = P + SI.

• CI: interest on interest; A = P(1 + r)^n.

• Compounding periods: yearly, half-yearly, quarterly.

• Effective rate of interest.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Simple and Compound Interest to activate prior knowledge.

2. Present the core ideas of Simple and Compound Interest with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Simple and Compound Interest.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Simple and Compound Interest and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Simple and Compound Interest for classroom display.

Assessment Questions

1. Find SI on Rs. 10,000 at 8% for 3 years.

2. Find CI on Rs. 10,000 at 10% for 2 years.

3. Which is greater after 3 years: SI or CI? Why?

Homework

• Write a one-page summary of Simple and Compound Interest in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)

Unit 8: Annuities: Present and Future Value

Learning Objectives

• Define annuity and its types.

• Find the future value of an ordinary annuity.

• Find the present value of an annuity.

Key Concepts

• Annuity: equal payments at regular intervals.

• Ordinary annuity vs annuity due.

• Future value: accumulated worth of payments.

• Present value: today's worth; used in loans.

Teaching Methodology (Step by Step)

1. Warm-up: ask students what they already know about Annuities: Present and Future Value to activate prior knowledge.

2. Present the core ideas of Annuities: Present and Future Value with clear board notes, definitions and real-life examples.

3. Explain each key concept in sequence, using questions to check understanding at every stage.

4. Guided practice: short written exercises and class discussion applying Annuities: Present and Future Value.

5. Recap the main points, clear remaining doubts and close with a quick oral quiz.

Classroom Activities

• Pair activity: students discuss the key ideas of Annuities: Present and Future Value and present a two-minute summary to the class.

• Group task: prepare a wall chart or mind map on Annuities: Present and Future Value for classroom display.

Assessment Questions

1. Define annuity. State its two types.

2. Find the future value of Rs. 1,000 yearly for 5 years at 6%.

3. A loan is repaid in instalments: which value concept applies?

Homework

• Write a one-page summary of Annuities: Present and Future Value in your notebook.

• Attempt the exercise questions given at the end of the chapter.

Time Allocation

2 periods (80 minutes)